Noushin Omidvar

Machine Learning Scientist | AI for Scientific Discovery

Santa Clara, California | Portfolio | LinkedIn | GitHub | Scholar

Profile

Machine learning scientist with a PhD in chemical engineering, developing predictive models and computational workflows for molecular and materials discovery. My work connects chemical representations, physical modeling, optimization, and experimental feedback to help scientists decide what to test next.

Professional Experience

Machine Learning Scientist, Aionics, Inc. **Apr 2024–Present**
Palo Alto, California

- Developed machine learning models for electrolyte mixture properties, including viscosity, ionic conductivity, and flash point. Combined molecular representations with formulation composition and test conditions to support screening and design.
- Built a closed-loop electrolyte discovery workflow combining mixture-property prediction, feasibility constraints, and multi-objective optimization. Used experimental feedback to revise formulation constraints and guide subsequent candidate selection.
- Modeled battery test outcomes using learning-to-rank for cycle life and mixture-based methods for gassing behavior, helping prioritize formulations and interpret experimental results.
- Built a molecular screening workflow for alternative working fluids, combining property prediction with chemical feasibility filters and thermodynamic calculations to identify candidates for engineering review.
- Evaluated pretrained molecular representations against descriptor-based baselines and developed reusable Python workflows for feature generation, model training, validation, and reporting.
- Collaborated with chemists and engineers to turn application requirements into modeling objectives, evaluate model limitations, and plan follow-up experiments.

Graduate Researcher, Virginia Tech **2016–2022**
Department of Chemical Engineering | Blacksburg, Virginia

- Developed TinNet, a physics-informed graph neural network linking atomic structure to electronic descriptors and Newns–Anderson chemisorption theory to predict adsorption energies and interpret catalyst reactivity.
- Combined DFT calculations, high-throughput screening, and active learning to identify promising oxygen-reduction catalyst surfaces and prioritize follow-up calculations.
- Applied interpretable modeling to connect surface bonding and electronic properties with catalytic activity, including studies of Pt-monolayer and alloy electrocatalysts.
- Collaborated with experimental researchers on catalyst design and mechanistic interpretation, including graph-learning work supporting the discovery of Ir-free ammonia-oxidation catalysts.

M.S. Researcher, Tarbiat Modares University **2012–2015**
Chemical Engineering | Tehran, Iran

- Led experimental research on biomaterial scaffolds for controlled drug delivery and bone tissue engineering, from microsphere preparation and electrospinning to characterization, release studies, stem-cell assays, analysis, and publication.

Technical Skills

Machine learning: scikit-learn, XGBoost, PyTorch, PyTorch Geometric, molecular property prediction, ranking, active learning, uncertainty-aware evaluation, multi-objective optimization.

Scientific computing: Python, RDKit, molecular descriptors and fingerprints, DFT (VASP), computational catalysis, numerical analysis.

Research software: Git, automated testing, data preparation and reproducible workflows, SQL, AWS/GCP, HPC/SLURM.

Education

Ph.D., Chemical Engineering, Virginia Tech 2022
M.S., Chemical Engineering, Tarbiat Modares University 2015
B.S., Chemical Engineering, Sharif University of Technology 2012

Selected Publications

- Omidvar, N., et al. "Interpretable machine learning of chemical bonding at solid surfaces." *The Journal of Physical Chemistry Letters*, 2021.
- Pillai, H. S., et al. "Interpretable design of Ir-free trimetallic electrocatalysts for ammonia oxidation with graph neural networks." *Nature Communications*, 2023.
- Omidvar, N., et al. "Explainable AI for optimizing oxygen reduction on Pt monolayer core–shell catalysts." *Electrochemical Science Advances*, 2024.
- Omidvar, N., et al. "In vitro osteogenic induction of human marrow-derived mesenchymal stem cells by PCL fibrous scaffolds containing dexamethasone-loaded chitosan microspheres." *Journal of Biomedical Materials Research Part A*, 2016.

Full publication list on Google Scholar